

CrewSight — Season business case

4-month season · 8 regattas · 32 tracking days · Handsets @ \$130.00 · IoT data \$10.00/SIM/month · Cloud infra included in net margin

IoT data rate for this variant: \$10.00/SIM/month (previous baseline \$5.00/SIM/month).

Profit summary (gross vs net of cloud)

Tier	Devices	Handset profit	Platform fee	Cloud cost	Platform (net)	Net profit	Revenue	Net margin %
50-device tier	50	\$2,195.50	\$6,000.00	\$127.75	\$5,872.25	\$8,067.75	\$14,500.00	55.6%
270-device tier	270	\$11,855.70	\$6,000.00	\$188.82	\$5,811.18	\$17,666.88	\$51,900.00	34.0%

Platform fee: \$6,000.00 flat per season (both tiers). Revenue column uses **data-included** pricing (4-month SIM). Net margin = handset margin + platform fee – cloud.

Cellular data — 4-month season vs alternatives (ex GST)

	50 devices	270 devices
Per SIM / month	\$10.00	\$10.00
Season months billed	4	4
Data included — full season (4 mo × \$10.00 × devices)	\$2,000.00	\$10,800.00
Regatta-only reference (32 days prorated — not customer default)	\$533.33	\$2,880.00
Data excluded — customer SIM (AHD cost)	\$0.00	\$0.00
Saving: excluded vs included (customer)	\$2,000.00	\$10,800.00

Customer pricing — data included vs excluded (ex GST)

	50 — included	50 — excluded	270 — included	270 — excluded
Handsets @ \$130.00	\$6,500.00	\$6,500.00	\$35,100.00	\$35,100.00
Cellular (4 mo)	\$2,000.00	—	\$10,800.00	—
CrewSight platform (season)	\$6,000.00	\$6,000.00	\$6,000.00	\$6,000.00
Season total	\$14,500.00	\$12,500.00	\$51,900.00	\$41,100.00
+ optional mounting @ \$50.00	\$17,000.00	\$15,000.00	\$65,400.00	\$54,600.00

Cloud & storage costs (season estimate)

Stack: **Vercel Pro** (AHD-LiveStream + rowing-app-recorder-pwa) · **Neon Postgres** (ingest/history) · **Upstash Redis** (CV positions, alert state) · Resend email (negligible).

50-device pilot

Vercel Pro (AHD-LiveStream + recorder ingest, 4 mo @ \$20)	\$80.00
Neon Postgres (~4.6 GB season samples @ \$5/mo + storage)	\$22.75

270-device fleet

Vercel Pro (AHD-LiveStream + recorder ingest, 4 mo @ \$20)	\$80.00
Neon Postgres (~24.8 GB season samples @ \$5/mo + storage)	\$34.86

Upstash Redis / KV (CV positions, alert state)	\$20.00
Bandwidth & CDN (maps, overlays)	\$5.00
Serverless overage (~10.9M ingest invocations, conservative)	\$0.00
Cloud subtotal (season)	\$127.75

~10.9M ingest samples · ~4.6 GB Postgres (season cumulative)

Upstash Redis / KV (CV positions, alert state)	\$40.00
Bandwidth & CDN (maps, overlays)	\$15.00
Serverless overage (~59.1M ingest invocations, conservative)	\$18.96
Cloud subtotal (season)	\$188.82

~59.1M ingest samples · ~24.8 GB Postgres (season cumulative)

Assumptions: 6,840 uploads/device/regatta day · ~450 B stored/sample (GPS, motion, events) · Vercel Pro \$20/mo × 4 months · Neon Launch \$5/mo + \$0.30/GB-month · Purge old sessions between regattas to control storage. Default 512 MB monitor quota will be exceeded at full fleet — budget Neon Launch and data management.

Handset economics

	Per unit	50 fleet	270 fleet
One NZ cost (Smart M26)	\$86.09	\$4,304.50	\$23,244.30
Customer sell price	\$130.00	\$6,500.00	\$35,100.00
Handset gross profit	\$43.91	\$2,195.50	\$11,855.70

50-device — data included

Handsets — cost (\$86.09 × 50)	\$4,304.50
Handsets — sell (\$130.00 × 50)	\$6,500.00
Handset margin (\$43.91/unit)	\$2,195.50
IoT data (4 mo @ \$10.00/SIM/mo × 50)	\$2,000.00
CrewSight platform (\$6,000.00 season fee)	\$8,000.00
Customer total (ex GST)	\$14,500.00
Customer total (incl GST)	\$16,675.00
Optional boat mounting (\$50.00 × 50)	\$2,500.00
With mounting (ex GST)	\$17,000.00

50-device — data excluded

Handsets — cost (\$86.09 × 50)	\$4,304.50
Handsets — sell (\$130.00 × 50)	\$6,500.00
Handset margin (\$43.91/unit)	\$2,195.50
IoT data (customer-provided — excluded)	\$0.00
CrewSight platform (\$6,000.00 season fee only)	\$6,000.00
Customer total (ex GST)	\$12,500.00
Customer total (incl GST)	\$14,375.00
Optional boat mounting (\$50.00 × 50)	\$2,500.00
With mounting (ex GST)	\$15,000.00

270-device — data included

Handsets — cost (\$86.09 × 270)	\$23,244.30
Handsets — sell (\$130.00 × 270)	\$35,100.00
Handset margin (\$43.91/unit)	\$11,855.70
IoT data (4 mo @ \$10.00/SIM/mo × 270)	\$10,800.00
CrewSight platform (\$6,000.00 season fee)	\$16,800.00
Customer total (ex GST)	\$51,900.00
Customer total (incl GST)	\$59,685.00
Optional boat mounting (\$50.00 × 270)	\$13,500.00
With mounting (ex GST)	\$65,400.00

270-device — data excluded

Handsets — cost (\$86.09 × 270)	\$23,244.30
Handsets — sell (\$130.00 × 270)	\$35,100.00
Handset margin (\$43.91/unit)	\$11,855.70
IoT data (customer-provided — excluded)	\$0.00
CrewSight platform (\$6,000.00 season fee only)	\$6,000.00
Customer total (ex GST)	\$41,100.00
Customer total (incl GST)	\$47,265.00
Optional boat mounting (\$50.00 × 270)	\$13,500.00
With mounting (ex GST)	\$54,600.00

Full delivery cost vs revenue (ex GST)

	50 incl.	50 excl.	270 incl.	270 excl.
Handset cost (One NZ)	\$4,304.50	\$4,304.50	\$23,244.30	\$23,244.30
IoT SIMs (4 months)	\$2,000.00	\$0.00	\$10,800.00	\$0.00
Cloud & storage (season)	\$127.75	\$127.75	\$188.82	\$188.82
Total delivery cost	\$6,432.25	\$4,432.25	\$34,233.12	\$23,433.12
Customer revenue	\$14,500.00	\$12,500.00	\$51,900.00	\$41,100.00
Net profit (after cloud + SIM)	\$8,067.75	\$8,067.75	\$17,666.88	\$17,666.88
+ optional mounting (\$50.00/unit)	\$2,500.00	\$2,500.00	\$13,500.00	\$13,500.00

Invoice structure

Invoice	50 incl.	50 excl.	270 incl.	270 excl.
Invoice 1 — Handsets (CAPEX)	\$7,475.00	\$7,475.00	\$40,365.00	\$40,365.00
Invoice 2 — CrewSight season (platform ± 4-mo data)	\$9,200.00	\$6,900.00	\$19,320.00	\$6,900.00
Optional — boat mounting	\$2,875.00	\$2,875.00	\$15,525.00	\$15,525.00

Recommendation: Quote **data included** for turnkey (\$59,685.00 incl GST, 270 fleet) or **data excluded** if customer has own SIMs (\$47,265.00 incl GST). Mounting optional at \$50.00/boat. Regatta-only SIM (\$2,880.00 internal) saves \$7,920.00 vs full season but is not the default customer offer.

Cellular & API load check

Cellular: ~267 MB/day (50) · ~1.44 GB/day (270) — within One NZ 500 MB/SIM pool. API: up to ~270 ingest POST/s during 1 s reporting hour (270 fleet). Serverless likely within Vercel Pro credits; overage modeled conservatively above.

M26 field validation (14 Jun 2026)

Field test 14 Jun 2026: 11:30–11:45 NZST drive (~10 km) at ~1 Hz. M26 median GPS accuracy **3.3 m** (p90 5.2 m) vs S21 reference 5.2 m; fix continuity 1 s max gap on M26. Track agreement median 4.2 m.

Battery 14 Jun 2026: 96% @ 06:45 NZ → 64% after 10.3 h (~3.1 %/h at 1 Hz GPS + background recording; ~32 h from full charge; app data 36 MB). An 8-hour active day at 1 Hz uses ~25% — no mid-day charge required.